

MAPA CORPORATE OPERATING SYSTEM

INVESTIGATION NAVIGATION MAP

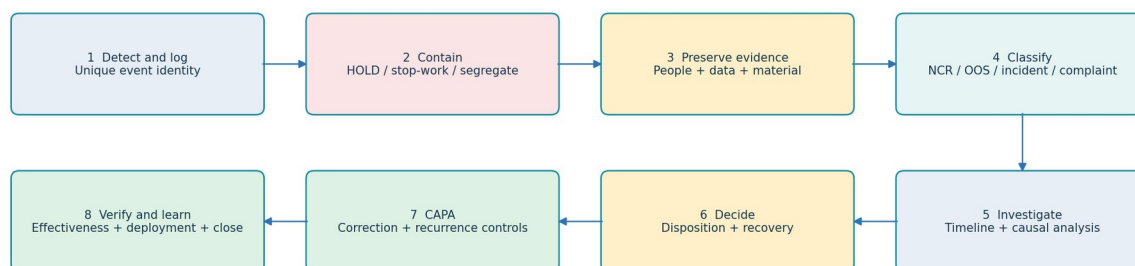
Contain, preserve, investigate, decide, correct and verify without losing traceability

Document code	MAPA-IMS-NAV-003
Gate / package	GATE-001 / Package-001 READ FIRST
Version / issue	1.0 / 08 August 2026
Owner	QA/QC & IMS Manager with relevant Technical / HSE / IT owner
Applicability	NCR, OOS, incident, complaint, failure, spill and data-integrity investigations
Status	Controlled Issue - GM activation approval required

INVESTIGATION RULE Recovery restores a controlled operating state; it does not, by itself, demonstrate that the root cause is removed or that CAPA is effective.

1. Investigation Route

Major investigation and CAPA route



Maintain chronology: actual activity time, discovery time, record entry time and decision time must remain distinct.

Figure 1 - Controlled investigation route applied in GATE-001.

2. Branch Selection

Branch	Trigger	Control path	Evidence to preserve	Decision owner
Product / batch / test result	OOS, OOT, atypical result, specification or genealogy concern	MAPA-IMS-PRO-NCR; MCS-QC-SOP-004 where applicable	Raw data, sample, method, equipment, analyst, calculation, retain sample	QA disposition / release authority

Branch	Trigger	Control path	Evidence to preserve	Decision owner
HSE / environmental event	Injury, near miss, unsafe condition, spill, emission or emergency	MAPA-HSE-PRO-EMR + MAPA-IMS-PRO-NCR	Scene, permit, LOTO, witness, exposure, spill/waste trace	HSE + accountable operations authority
IT/OT / data integrity	Access, outage, record loss, interface, backup, restore or trusted-state concern	MAPA-IT-REG-004; IT-FRM-005/006; IT-CHK-004; MAPA-IMS-PRO-NCR	Audit trail, logs, backup, hash, interfaces, reconciliation and RTS evidence	IT/OT + process owner + independent verifier
Supplier / warehouse	Unapproved source, receipt, identity, condition, segregation or preservation failure	MAPA-SCM-PRO-SUP + warehouse records + MAPA-IMS-PRO-NCR	PO, approved source, CoA, receipt, inspection, status and custody	SCM / warehouse / QA
Customer / service / report	Complaint, field failure, service deviation or amended report	Customer process + MAPA-IMS-PRO-NCR + release control	Contract, requirement, report/version, field data and communication	Commercial /technical owner + QA

3. Major Investigation Minimum Content

1. Define the event statement without embedding an assumed cause.
2. Freeze the affected scope, status and evidence; record what is known, unknown and excluded.
3. Build a time-sequenced chronology using source timestamps and named actors.
4. Test plausible causes across people, method, material, equipment, environment, data and management system.
5. Distinguish direct cause, contributing factors, root/system cause and failed or missing barriers.
6. Assess product/service disposition, HSE/environmental consequences, data integrity and legal/customer notification.
7. Define corrections and CAPA with owner, due date, acceptance criteria, operating exposure and independent verifier.
8. Verify effectiveness against recurrence and adverse-effect criteria; deploy learning horizontally where applicable.

4. GATE-001 Applied Investigation Record

Element	Record	Controlled statement	Status
Event	INCIDENT-001	Integrated operational/data-integrity recovery scenario	Closed within simulation gate
Classification	Major Investigation / CAPA	Cross-functional evidence and independent recovery verification required	Accepted
Containment	HOLD and scope preservation	No live release before trusted-state and reconciliation checks	Verified
Recovery	Trusted recovery + return-to-service reconciliation	Recovery evidence accepted after functional/data/interface checks	Accepted

Element	Record	Controlled statement	Status
CAPA	Recurrence controls + horizontal deployment	Effectiveness remains separate from recovery acceptance	Owned / monitored
Gate	GATE-001	Eight-part case file and decision chain retrievable	PASS

5. No-Go and Re-Open Triggers

- Unknown affected scope, lost chronology, altered original record or missing source evidence.
- Recovery point is not demonstrably clean/trusted, or interfaces cannot be reconciled.
- Product, service, HSE/environmental or regulatory disposition remains open.
- A critical action has no owner, due date, acceptance criteria or independent verifier.
- Recurrence, adverse effect, failed effectiveness check or new contradictory evidence appears.

6. Closure Test

Closure domain	PASS evidence	No-go
Containment	Effective and evidence preserved	No uncontrolled exposure
Causal analysis	Evidence supports cause and barrier conclusions	No unsupported single-cause closure
Disposition/recovery	Authorized decision with retrievable acceptance evidence	No release by the executor alone
CAPA	Actions match causes; residual risk accepted	No administrative-only fix for engineering/system cause
Effectiveness	Predetermined criteria and adequate operating exposure	Recovery success is not effectiveness proof
Learning	Trend, lesson and horizontal deployment recorded	No isolated local closure where systemic

Revision History

Version	Date	Change	Authority / record
0.1	06.08.2026	GATE-001 simulation working content	Simulation baseline
0.9	07.08.2026	Gate closure; navigation and case route frozen	GATE-001 PASS
1.0	08.08.2026	Initial controlled Document Package issue	CN-2026-0001

Approval and Activation

Stage	Role	Decision condition	Status / date
Prepared	QA/QC & IMS Manager	Content and traceability complete	08.08.2026
Reviewed	Technical Director / Functional Owners	Controlled review required	Open
Approved	General Manager	Required before live operational activation	Pending

ACTIVATION BOUNDARY This approval page authorizes live operational use only when the required functional review and General Manager decision are completed in the controlled source record.